

① 5 ลูก 2 สี 3

$$\Omega = 5 \times 5 = 25$$

$$P(X < 1)$$

คือ สีฟ้ากับสีชมพู = $9/25$

$$P(X < 1) = 1 - \left(\frac{4}{25} + \frac{9}{25} \right)$$

$$= 1 - \frac{13}{25}$$

$$= \frac{12}{25} \quad \#$$

$$\text{สีฟ้า} \quad 2 \times 2 = 4/25$$

$$\text{สีชมพู} \quad 3 \times 3 = 9/25$$

$$\text{สีอื่น} \quad \frac{25-13}{25} = 12/25$$

$$p_X(X) = \begin{cases} 4/25, & X = 2 \\ 9/25, & X = 3 \\ 12/25, & X = -1 \\ 0, & \text{otherwise} \end{cases}$$

② $E[X]$

$$E[X] = \left(2 \times \frac{4}{25} \right) + \left(3 \times \frac{9}{25} \right) + \left(-1 \times \frac{12}{25} \right)$$

$$= \frac{8}{25} + \frac{27}{25} + \frac{-12}{25}$$

$$= \frac{23}{25} \quad \#$$

③ $\text{Var}[X] \rightarrow E[X^2] - (E[X])^2 \quad | \quad E[X] = \frac{23}{25}$

$$E[X^2] = \left(2^2 \times \frac{4}{25} \right) + \left(3^2 \times \frac{9}{25} \right) + \left((-1)^2 \times \frac{12}{25} \right)$$

$$= \frac{16}{25} + \frac{81}{25} + \frac{12}{25}$$

$$= \frac{109}{25}$$

$$0.0 \quad \frac{109}{25} - \left(\frac{23}{25} \right)^2 =$$

$$\frac{109}{25} - \frac{529}{625}$$

$$\frac{2725}{625} - \frac{529}{625} = \frac{2196}{625} \quad \#$$